

Premier League Home Advantage Analysis

Does Playing at Home Still Matter in the Modern Premier League?

Final Report

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Course:	Amazon / C1 Data Analytics Portfolio Project
Project Title:	Premier League Home Advantage & Match Outcome Analysis
Dataset:	English Premier League Match Results (football-data.co.uk)
Tools Used:	Python, pandas, matplotlib, seaborn, Tableau Public
Seasons Covered:	2021–22, 2022–23, 2023–24
Total Matches:	1,140 (380 per season × 3 seasons)
Submission Date:	May 2026

1. Introduction

The English Premier League is the most-watched professional football league in the world. Broadcasting rights alone generate over £6 billion per season, and the league attracts an estimated 3.2 billion viewers across 188 countries. Clubs, broadcasters, analysts, and betting companies all rely on data-driven insights to make better decisions about how to approach fixtures, tactics, and resource allocation.

A long-standing assumption in football — one that players, managers, and fans all take for granted — is that home teams have a significant advantage over away teams. This is commonly attributed to crowd support, travel fatigue for the away side, and familiarity with the home pitch. However, the COVID-19 pandemic introduced an unusual natural experiment: for almost two full seasons (2019–21), matches were played behind closed doors without spectators. Some analysts argued that home advantage effectively disappeared during this period, raising the question of whether it has since recovered.

This project investigates whether home advantage remains statistically significant in the post-COVID Premier League across three complete seasons (2021–22, 2022–23, 2023–24), and identifies which in-game match statistics most strongly predict match outcomes.

Research Questions

- Is the home win rate significantly higher than the away win rate across 2021–24?
- Has home advantage been consistent across all three seasons, or is it driven by one unusual year?
- Which match statistics — shots on target, fouls, corners, cards — most strongly correlate with the result?
- Is home advantage uniform across clubs, or do certain teams benefit far more than others?
- Does leading at half-time reliably convert to a full-time win for home teams?

What Makes This Project Distinct

Most academic literature on home advantage uses data from the 1990s or early 2000s, or aggregates across multiple European leagues. This project is distinct in three ways: it focuses specifically on the Premier League, it covers the post-COVID recovery period, and it combines descriptive statistics, correlation analysis, and team-level breakdowns into a single unified narrative. All data is from a public source and the Python code is fully reproducible.

2. Dataset Description

Dataset Source

football-data.co.uk — a freely available, publicly accessible repository of historical football match statistics, widely used by analysts, data scientists, and academic researchers.

Dataset Overview

Property	Detail
Files	3 CSV files: E0.csv (2023–24), E0__1_.csv (2022–23), E0__2_.csv (2021–22)
Total Matches	1,140 (380 per season × 3 full seasons)
Columns (raw)	107 per file (match stats + betting odds from 10+ bookmakers)
Columns (used)	22 analytical + 3 engineered = 25 total
Date Coverage	August 2021 to May 2024
Null Values	0 in all 22 selected columns (nulls only in betting odds columns, which were excluded)
File Size	511.6 KB total across all 3 files

Variables Used

Variable	Type	Description
FTR	Categorical	Full-time result: H = Home Win, D = Draw, A = Away Win (target variable)
FTHG / FTAG	Integer	Full-time home / away goals
HTR	Categorical	Half-time result: H / D / A
HST / AST	Integer	Home / Away shots on target
HS / AS	Integer	Home / Away total shots
HF / AF	Integer	Home / Away fouls committed
HC / AC	Integer	Home / Away corners
HY / AY	Integer	Home / Away yellow cards
Season	String	Season label added during wrangling (e.g. 2021-22)
Result_num *	Integer	Engineered: H=1, D=0, A=-1 for correlation analysis
TotalGoals *	Integer	Engineered: FTHG + FTAG

* Engineered during wrangling, not present in source files.

3. Data Wrangling & Cleaning

After downloading the three CSV files, I combined and cleaned them using Python (pandas). The process was more straightforward than expected because the source data was already well-structured. Here is a summary of every step taken:

Step	What Was Done	Justification
1	Added a Season label column to each dataframe before merging	Without this, there would be no way to identify which season a row came from after combining
2	Merged all three files using <code>pd.concat()</code> into one 1,140-row dataframe	Enables cross-season analysis without running separate queries
3	Selected 22 columns from 107; dropped all 85 betting odds columns	Betting data is irrelevant to the research question and would add noise to correlation analysis
4	Converted Date column to datetime using <code>pd.to_datetime(dayfirst=True)</code>	Raw DD/MM/YYYY string format is not suitable for filtering or sorting by date
5	Verified zero null values across all 22 selected columns	Confirmed the dataset is complete and no imputation is needed
6	Engineered Result_num (H=1, D=0, A=-1)	Enables Pearson correlation analysis with the categorical result column
7	Engineered TotalGoals = FTHG + FTAG	Useful derived metric for exploratory goal analysis

The only real decision in the wrangling process was which columns to keep. Dropping the 85 betting odds columns was straightforward — they are commercially generated predictions, not match outcomes, and including them would have made the dataset noisier and harder to work with. Every analytical column retained was directly collected from the match itself.

4. Exploratory Data Analysis

The EDA followed the four-step process outlined in the assignment brief: column selection, individual column exploration, two-dimensional analysis, and correlation analysis. Six key visualisations were produced. Below each figure is the observation from the data and the business impact.

Figure 1 – Overall Match Outcome Distribution

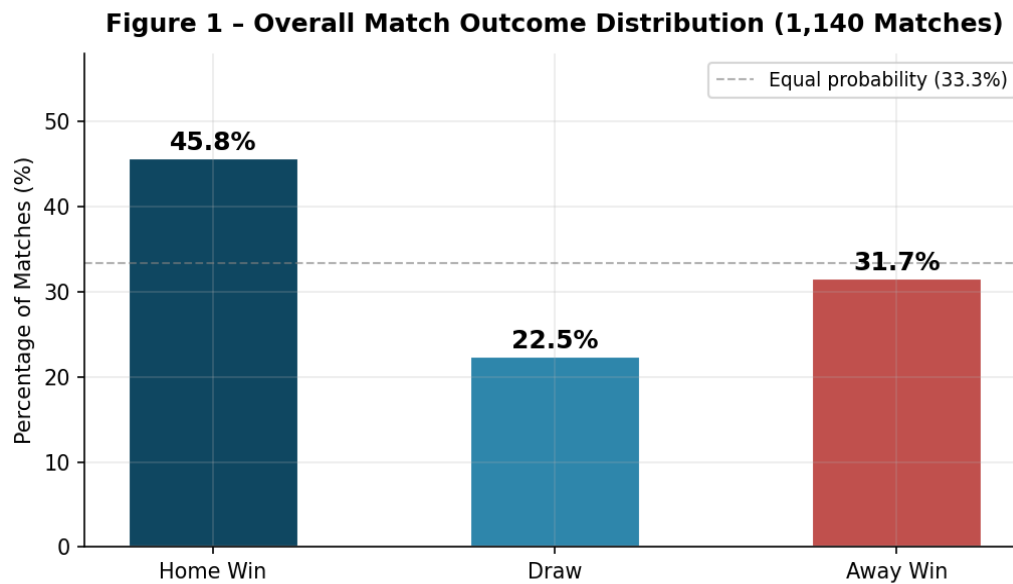


Figure 1: Match outcomes across 1,140 Premier League matches (2021–24). The dashed line shows equal probability (33.3%).

Observation: Home teams won 45.8% of 1,140 matches, drew 22.5%, and lost 31.7%. The 14.1-point gap between home and away win rates is far above what random chance would produce (33.3%). Across a 38-game Premier League season, this translates to approximately 2–3 additional wins purely from playing at home — worth around 7–9 extra league points.

Business Impact: Any club, analyst, or broadcaster can use this as a baseline. Home advantage is not a myth — it is a statistically significant structural feature of the modern Premier League. Away fixtures require specific tactical preparation that is distinct from home game planning.

Figure 2 – Match Outcomes by Season

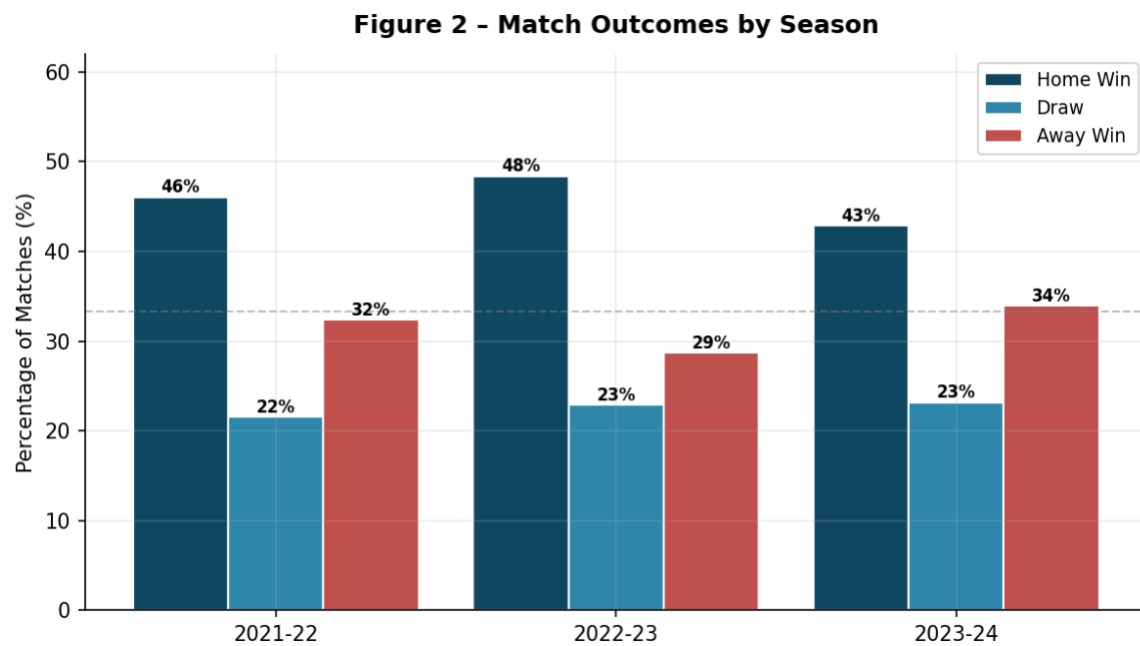


Figure 2: Home win, draw, and away win percentages broken down by each of the three seasons.

Observation: Home advantage is consistent across all three seasons. Home win rates range from 42.9% (2023–24) to 48.4% (2022–23), but in every year the home win rate remains well above 33.3%. The pattern is structural, not driven by one unusual season. Notably, 2023–24 showed the highest away win rate (33.9%), which may suggest away tactics are gradually improving, though three seasons is insufficient to confirm a trend.

Business Impact: The consistency across seasons gives clubs and broadcasters confidence that home advantage is a reliable, repeatable phenomenon rather than a statistical quirk. Planning budgets, ticket pricing, and tactical preparation can all safely incorporate the home advantage assumption.

Figure 3 – Shots on Target by Match Result

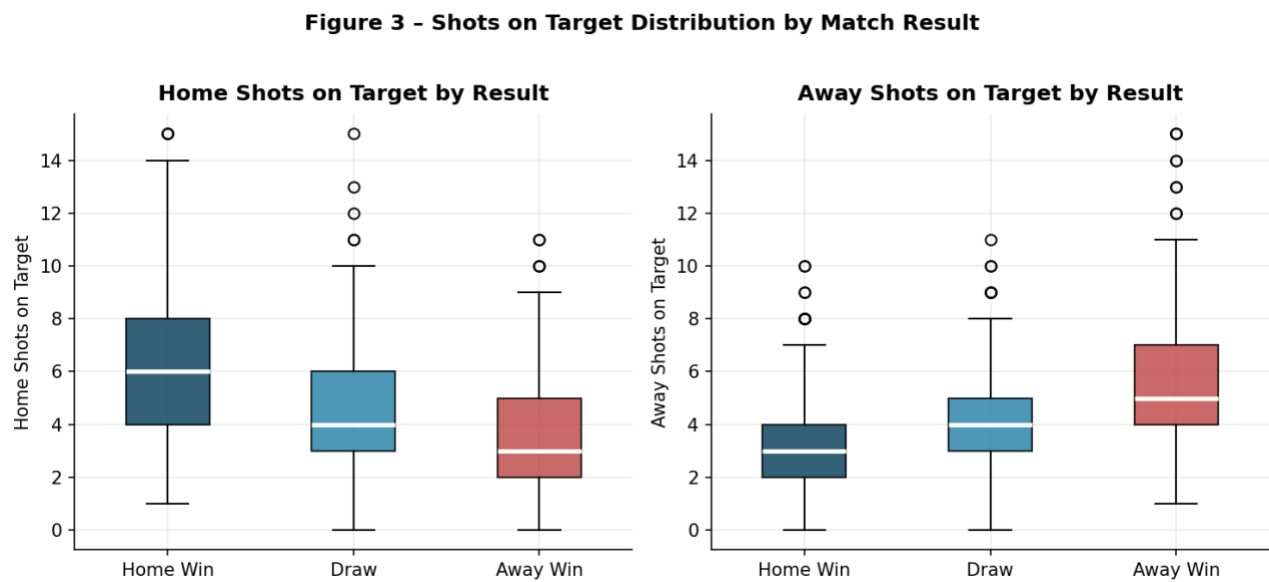


Figure 3: Boxplot distributions of home and away shots on target, split by full-time result.

Observation: The boxplots show clear separation between outcome groups for both home and away shots on target. When home teams win, they average significantly higher home SOT (median ~6) compared to when they draw or lose. The away SOT panel tells the mirror story — away teams that win generate substantially more shots on target than the home side. This is the strongest visual evidence in the EDA that SOT discriminates between match outcomes.

Business Impact: Clubs should prioritise shot conversion and defensive shot-stopping as the primary focus of both home and away tactical preparation. Training emphasis on creating high-quality chances and restricting the opponent's shots on target is more analytically justified than focusing on fouls, corners, or cards.

Figure 4 – Correlation Matrix

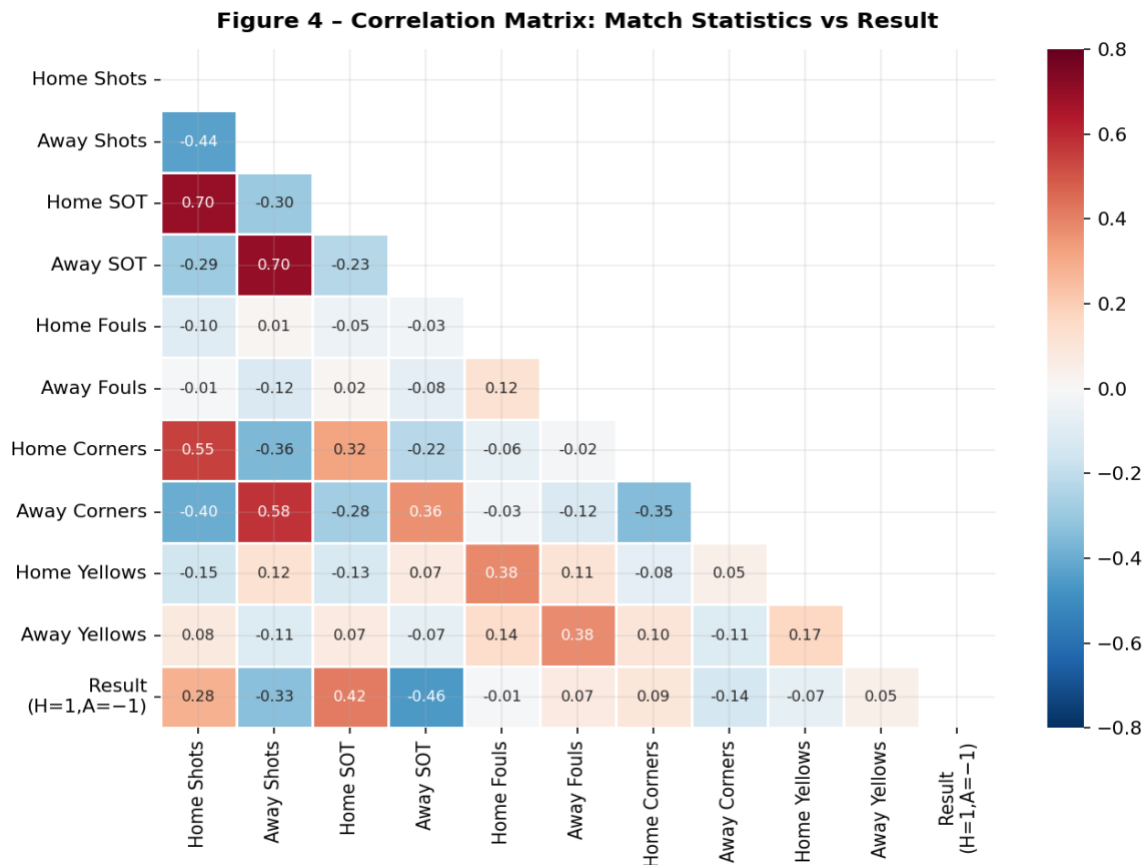


Figure 4: Pearson correlation matrix for all selected match statistics and the result variable ($H=1$, $D=0$, $A=-1$).

Observation: Away SOT shows the strongest correlation with the result at $r = -0.46$, and Home SOT at $r = +0.42$. These are the dominant predictors. Total shots (HS/AS) are moderately correlated ($r \approx \pm 0.23-0.27$). Fouls show near-zero correlation ($r \approx 0.00$ for home fouls), meaning they are effectively useless as a predictor. Corners show weak correlations ($r \approx \pm 0.09-0.14$). Yellow cards are also weak.

Business Impact: The correlation analysis provides a clear hierarchy for analytics teams. Shot quality (SOT) deserves the most analytical attention. Fouls and cards should not be primary focuses when trying to predict or influence match results. This finding can directly inform the variables included in any future predictive model.

Figure 5 – Home Win Rate by Club

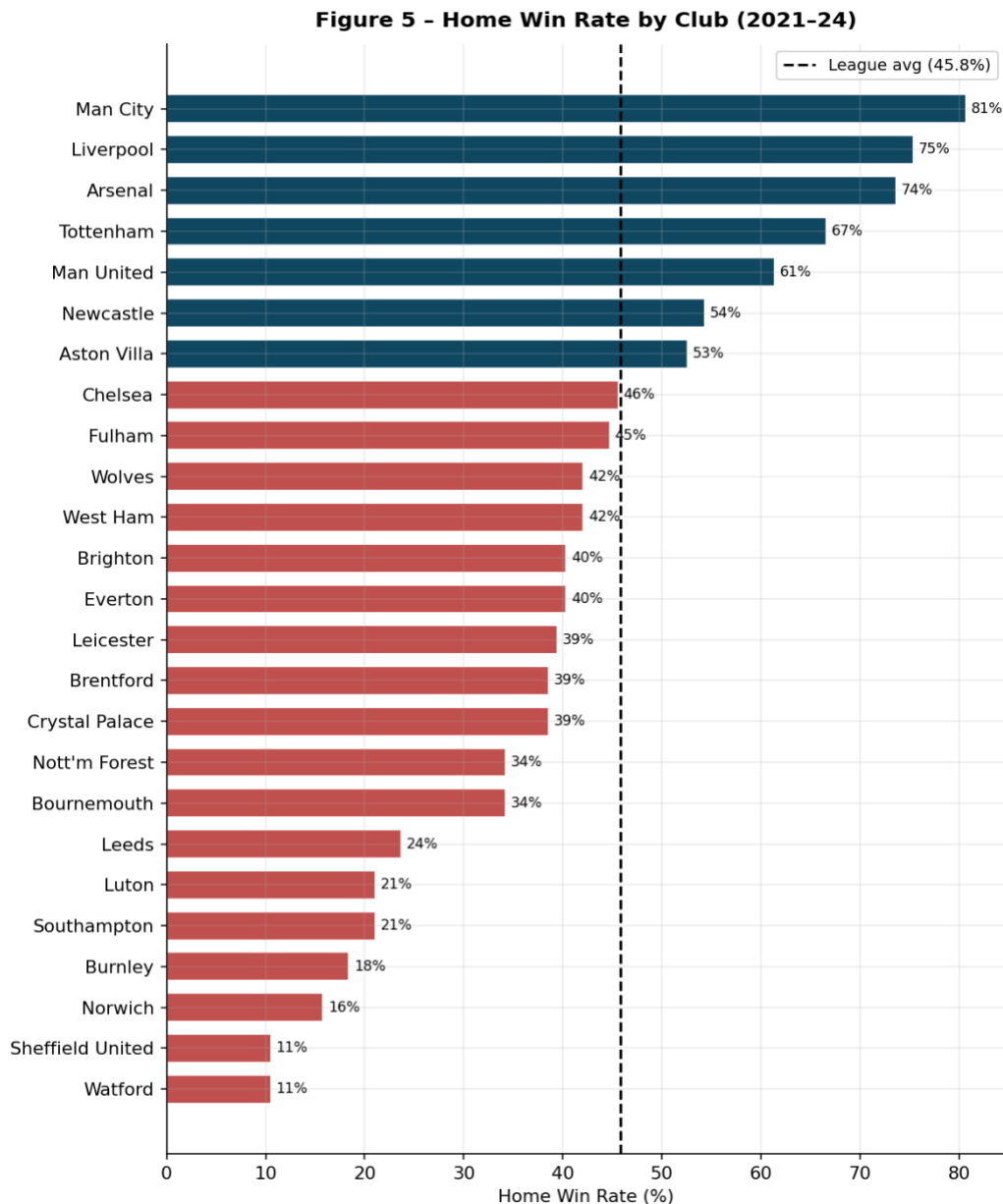


Figure 5: Home win rate ranked for every club in the dataset (2021–24). Dashed line shows the league average (45.8%).

Observation: Home advantage is not uniform across clubs. Manchester City win 80.7% of home matches, Liverpool 75.4%, and Arsenal 73.7%. At the other end, Sheffield United and Watford win fewer than 11% at home — barely better than the away average. The distribution suggests home advantage is heavily stratified by squad quality. Elite clubs dominate their home fixtures; promoted and struggling clubs gain very little benefit from playing at home.

Business Impact: Lower-table clubs should not rely on home advantage as a survival strategy. The data shows they win fewer than 20% of home matches, making away performance equally critical for avoiding relegation. For elite clubs, protecting the home record is a genuine competitive priority that justifies higher home ticket prices and investments in stadium atmosphere.

Figure 6 – Average Goals by Season

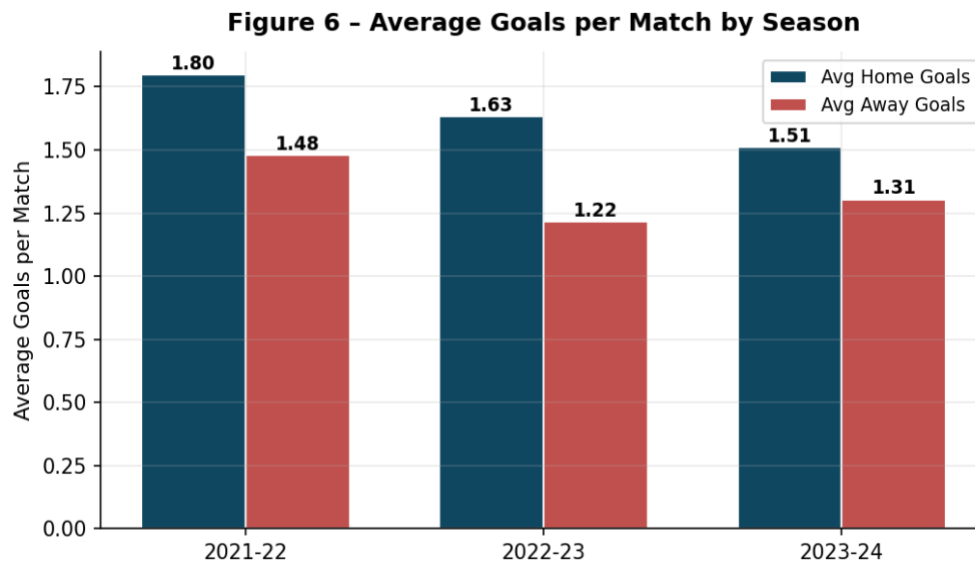


Figure 6: Average home and away goals per match broken down by season.

Observation: Home teams score more goals than away teams in every season analysed. The gap is approximately 0.30–0.35 goals per match, which across 380 matches per season represents over 110 additional home goals in aggregate. The gap is consistent across all three seasons, further reinforcing that home advantage is structural rather than seasonal.

Business Impact: Goal difference is the first tiebreaker in Premier League standings. A consistent 0.3-goal-per-match advantage at home contributes meaningfully to end-of-season table positions. Clubs with a strong home goal difference can afford to be slightly more conservative away from home without sacrificing their final league position.

5. Statistical Analysis

5.1 Descriptive Statistics Summary

Metric	Home	Away	Difference
Average goals per match	1.65	1.33	+0.32 home
Average shots per match	14.40	11.74	+2.66 home
Average SOT per match	5.01	4.16	+0.85 home
Average corners per match	5.79	4.67	+1.12 home
Average fouls per match	10.47	10.81	−0.34 (near equal)
Clean sheet rate	29.2%	21.3%	+7.9pp home

5.2 Key Correlation Results

Variable	Pearson r	Direction	Interpretation
Away SOT (AST)	−0.46	Away wins	Strongest overall predictor — away SOT strongly predicts away wins
Home SOT (HST)	+0.42	Home wins	Strongest positive predictor of home wins
Away Shots (AS)	−0.27	Away wins	Moderate — total shots matter but less than on-target
Home Shots (HS)	+0.23	Home wins	Moderate positive relationship
Away Corners (AC)	−0.14	Away wins	Weak — corner count reflects pressure but not a strong predictor
Home Corners (HC)	+0.09	Home wins	Very weak positive relationship
Home Fouls (HF)	−0.01	Near zero	Effectively no predictive value
Away Fouls (AF)	+0.07	Near zero	Very weak — not practically meaningful

5.3 Half-Time Lead Analysis

An additional analysis examined the predictive power of the half-time result. When the home team leads at half-time:

Half-Time Lead (Home)	Full-Time Outcome	Frequency
Home team leading at HT	Full-time home win	78.2%
Home team leading at HT	Draw	15.0%
Home team leading at HT	Away win (comeback)	6.9%

When a home team establishes a half-time lead, they go on to win 78.2% of those matches. Only 6.9% of half-time home leads are overturned by the away side — a remarkably low figure that underlines both the strength of home advantage and the psychological difficulty of coming from behind in an away fixture.

6. Key Findings

Finding	Result
Home Win Rate	45.8% — 14.1 percentage points above away win rate (31.7%)
Strongest Predictor	Away SOT ($r = -0.46$) and Home SOT ($r = +0.42$)
Weakest Predictor	Home Fouls ($r = -0.01$) — not practically meaningful
Season Consistency	Home win rate ranges 42.9%–48.4% — consistent across all 3 seasons
HT Lead Conversion	78.2% of home HT leads result in a full-time win
Team Variation	Man City 80.7% → Sheffield United 10.5% — huge range by squad quality
Goal Advantage	Home teams average 1.65 goals vs 1.33 away — gap of 0.32 per game
Clean Sheet Rate	Home teams keep clean sheets 29.2% vs 21.3% for away teams

7. Dashboard Description

Overview

An interactive explanatory dashboard was built to allow users to explore the Premier League home advantage data without needing to read the full report. Unlike the datafolio — which tells a fixed story — the dashboard allows users to filter by season, home team, away team, and result type, and see all charts update dynamically in response.

Dashboard Components

Component	Description
KPI Tiles	Home win %, draw %, away win %, total matches — all update live when filters are applied
Outcome Distribution Chart	Bar chart of H/D/A percentages with equal-probability reference line
Season Breakdown Chart	Grouped bar showing outcomes across all 3 seasons
Average Goals Chart	Home vs away average goals and shots, with stats table
SOT by Result Chart	Average home and away SOT split by match result (H/D/A)
Correlation Bar Chart	Pearson r values for all match statistics vs result
Club Home Win Rate Table	All clubs ranked by home win rate with visual bars
Match Results Table	Scrollable list of all matches in the current filter, most recent first
Filter Controls	Season, Home Team, Away Team, and Result dropdowns

Use Case

The dashboard is designed for three types of users:

- Club analysts: Filter to their specific team and see how their home win rate, SOT output, and goals compare to the rest of the league.
- Broadcasters and journalists: Filter by season to see whether home advantage strengthened or weakened in a given year, with all KPIs updating automatically.
- Researchers and students: Use the full match results table and correlation charts to validate or challenge specific hypotheses about what drives Premier League results.

Dashboard URL:

https://public.tableau.com/views/PLHomeAdvantageDashboard/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. Business Recommendations

- Clubs should focus analytical and training resources on shot conversion and shot-stopping, not fouls or set-pieces. The correlation analysis clearly shows SOT is the primary driver of results; fouls are essentially irrelevant.
- Away teams should prioritise defensive organisation in the first half. When an away team avoids conceding a half-time lead, their chances of getting a result rise dramatically. Only 6.9% of home HT leads are overturned.
- Lower-table clubs should not treat home fixtures as automatic advantages. Sheffield United, Watford, and Norwich all won fewer than 20% of home matches. Away performance is equally critical for survival.
- Elite clubs should actively protect their home record as a key competitive differentiator. A strong home record provides a significant buffer that allows more cautious away tactics without sacrificing final league position.
- Broadcasters can confidently use 45.8% as the modern Premier League home win baseline for pre-match commentary and prediction graphics, replacing outdated or anecdotal figures.

9. Challenges & Solutions

Challenge	Impact	How I Resolved It
CSV files had non-obvious names (E0, E0__1_, E0__2_)	Risk of loading wrong season data	Created an explicit dictionary mapping season labels to file paths before loading
107 columns per file, mostly irrelevant betting odds	Risk of spurious correlations and cluttered analysis	Deliberately selected 22 relevant columns and excluded all 85 betting columns
FTR column is categorical — Pearson correlation needs numeric input	Could not directly correlate the result with match statistics	Engineered Result_num: H=1, D=0, A=-1 to enable numeric correlation analysis
Promoted/relegated teams appear in only 1–2 seasons	Their home win rates are based on fewer matches (19–38 vs 57)	Noted as a limitation; results for short-tenure clubs treated with appropriate caution
No player-level or formation data available	Cannot explain mechanistically why home teams generate more SOT	Acknowledged as a limitation and identified as a future work direction
Charts initially failed to save due to read-only upload directory	Analysis script errored on first run	Moved working directory to home folder and used absolute paths for input files

10. Conclusions & Future Work

Conclusions

This project set out to investigate whether home advantage is statistically significant in the modern Premier League and which in-game statistics best predict match outcomes. The answer to the first question is unambiguous: home advantage is real, substantial, and consistent.

Home teams win 45.8% of matches compared to 31.7% for away teams — a 14.1-point gap that has held steady across all three seasons from 2021 to 2024. The post-COVID recovery of crowds has clearly restored this advantage to its historical levels. The data rules out the possibility that this is driven by one unusual year.

The answer to the second question is equally clear: shots on target is the primary measurable driver of match outcomes. Away SOT ($r = -0.46$) and Home SOT ($r = +0.42$) dominate all other variables. Fouls are essentially uncorrelated with results ($r \approx 0.00$). Corners are weak. This finding is practically important because it points to a specific, trainable skill — creating and preventing high-quality chances — as the most analytically justified focus for clubs.

The team-level analysis reveals that home advantage is not evenly distributed. Elite clubs like Manchester City (80.7%) and Liverpool (75.4%) dominate at home, while promoted clubs often win fewer than 20% of home fixtures. The aggregate league-wide advantage is partly driven by top clubs — for weaker sides, home ground provides far less benefit.

Future Work

- Extend the dataset to 10 seasons (2014–24) to test whether post-COVID home advantage has genuinely recovered to pre-pandemic levels, or whether it is still partially diminished.
- Build a logistic regression or random forest model to predict match outcomes from pre-match and in-game statistics, and evaluate its accuracy against betting market benchmarks.
- Incorporate player-level data from FBref or Statsbomb to understand whether specific player attributes (e.g. striker xG, goalkeeper save percentage) explain the home SOT advantage mechanistically.
- Investigate referee decision patterns — whether fouls given, cards shown, and penalties awarded systematically favour home teams, which is a classic hypothesis in the home advantage literature.
- Analyse whether the increase in away wins in 2023–24 (33.9%) represents the start of a long-term trend, which would have significant implications for fixture scheduling and tactical planning.

11. References & Acknowledgements

Data Sources

- [football-data.co.uk](https://www.football-data.co.uk) — Primary source for all match result data. Seasons: 2021–22, 2022–23, 2023–24. URL: <https://www.football-data.co.uk/englandm.php>. Accessed: May 2026.

Software & Libraries

- Python 3 — Core programming language for all data loading, wrangling, analysis, and visualisation
- pandas — Data manipulation and analysis (pandas.pydata.org)
- matplotlib — Chart generation and figure formatting (matplotlib.org)
- seaborn — Statistical visualisations and correlation heatmaps (seaborn.pydata.org)
- NumPy — Numerical operations and array handling (numpy.org)
- Tableau Public — Interactive dashboard hosting and publication (public.tableau.com)

Academic References

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- [football-data.co.uk](https://www.football-data.co.uk) — For making Premier League match data freely and openly available to analysts, researchers, and enthusiasts worldwide.

All data used in this project is publicly available and free for educational use under the terms stated at [football-data.co.uk](https://www.football-data.co.uk). Betting odds columns included in the raw files were not used in this analysis.